

Module 4 Assignment II

Student A: Diya Jain

Student B: Suraj Kumar

Section I

Code by Diya Jain

Comments by Suraj Kumar

Final Answer

1. A. addi t0, x0, zero

#i=0

For:

 bge t0, a1, break

#if statement

 slli t1, t2, 2

i*4

 lw t3, t1(a0)

#temp[i]=0

 slli t3, t3, 7

temp[i]*128

 sw t3, t1(a0)

#storing to memory

 addi t0, t0, 1

#incrementing

 blt t0, a1, for

calling loop

B. addi t0, x0, 0

#a=0

 addi t1, x0, 1

#b=1

 sw t0, 0(a0)

#fib[0] = a

 sw t1, 4(a0)

#fib[1] = b

 addi t2, x0, 2

i=2

for:

 bgt t2, a1, exit

#if condition

 add t5, t0, t1

c= a+b

 addi t0, t1, 0

a=b

 addi t1, t5, 0

b=c

 slli t3, t2, 2

t3 = i*4

 add t3, t3, a0

t3= address of fib[i]

 sw t5, 0(t3)

#fib[i] =c

 addi t2, t2, 1

#i++

 j for

#calling for

exit:

C. addi t0, x0, 0

i=0

while:

 add t1, t0, a0

address of str[i] = t0

 lb t2 0(t1)

t1 = str[i]

 add t3, t1, a1

 begz t2, exit

if:	addi t3, x0, 'a'	# t3 = zero + 'a'
	addi t5, x0, 'z'	# t5 = zero + 'z'
	blt t2, t4, elif	# branch t2 < t4 and then jump to elif
	bgt t2, t5, elif	# second condition of if statement. t2 > t5
		then jump to elif
	sub t2, t5, t4	# t2 = t5 - t4
	sub t2, t5, t2	#t2 = t5 - t2
	sb t2, 0(t3)	#store byte in t2 from 0 t3
	J end	
elif:		
	addi t4, x0, 'A'	# t4= zero + 'A'
	addi t5, x0, 'Z'	#t5= zero + 'Z'
	blt t2, t4, else	#branch less than statement
	blt t2, t5, else	#another branch less than statement
	sub t2, t2, t4	# t2= t2-t4
	sub t2,t5,t2	#t2= t5-t2
	sb t2, 0(t2)	# store byte from t2
	J end	# jump
else:		
	sb t2, 0(t3)	# store byte from t2
end:		
	addi t1, t1, 1	# t1= t1 +1
	j while	# jump
exit:		

Judgement: Done. Code is clear

Discussion: This question has if, while and for loops. We used loading and storing to access the array elements.

Q2-)

DRAFT

#nums = a0, size = a1

#target = a2

Linear Search:

addi	t1, a1, -1	#t1= a1
slli	t1, t1, 2	#t1= t1*2
add	t0, a0, t1	# t0= a0 *t1
lw	t2, 0(a0)	# load word

while:

bgt	a0, t0, if	# if a0 > t0 jump to if
beq	t2, a2, if	# branch equal to
addi	a0, a0, 4	# a0= a0+4

```

        j while                #jump
if:
    bgt    a0, t0, exit        #branch greater than
    sub    t3, t0, a0          # subtraction
    srai   t3, t3, 2           # t3= t3/ 2^2
    addi   a0,t3,1             # a0 = t3 +1
    jr ra                      #jump
exit:
    addi   a0, a0, -1          #a0= a0 -1
    jr ra                      #jump
done:

```

Judgement: Code is clear and easy to understand.

Discussion: This question had while and if statement that had to be written.

Final Answer:

#nums = a0, size = a1

#target = a2

Linear Search:

```

    addi   t1, a1, -1          #t1= a1
    slli   t1, t1, 2           #t1= t1*2
    add    t0, a0, t1          # t0= a0 *t1
    lw     t2, 0(a0)           # load word

```

while:

```

    bgt    a0, t0, if          # if a0 > t0 jump to if
    beq    t2, a2, if          # branch equal to
    addi   a0, a0, 4           # a0= a0+4
    j while                #jump

```

if:

```

    bgt    a0, t0, exit        #branch greater than
    sub    t3, t0, a0          # subtraction
    srai   t3, t3, 2           # t3= t3/ 2^2
    addi   a0,t3,1             # a0 = t3 +1
    jr ra                      #jump

```

exit:

```

    addi   a0, a0, -1          #a0= a0 -1
    jr ra                      #jump

```

done:

Q3-)

DRAFT

Add	t0,x0,x0	#t0= x0+x0
Addi	t1,a1,-1	#t1=a1-1
While:		
bgt	t0,t1,end	#branch greater than
sub	t2,t1,t0	#t2= t1-t0
srli	t2, t2, 1	#t2= t2/2
add	t2,t0,t2	
if:		
slli	t3,t2,2	# t3= t2*4
add	t3,a0,t3	#t3= a0+t3
lw	t4,0(t3)	#loadword
bne	a2,t4,elif	#if a2!=t4 jump to elif
add	a0,x0,t2	# a0=x0+t2
jr	ra	#jump
elif:		
bge	a2,t4,else	#if a2>t4 jump to elif
addi	t1,t2,-1	# t1= t2-1
j while		#jump
else:		
addi	t0,t2,1	#t0=t2+1
j while		#jump
end:		
addi	a0, x0, -1	
jr	ra	

Judgement: Code is clear and concise

Discussion Summary: This question was a little tough to do because it had if and while together

Final Answer:

Add	t0,x0,x0	#t0= x0+x0
Addi	t1,a1,-1	#t1=a1-1
While:		
bgt	t0,t1,end	#branch greater than
sub	t2,t1,t0	#t2= t1-t0
srli	t2, t2, 1	#t2= t2/2
add	t2,t0,t2	
if:		
slli	t3,t2,2	# t3= t2*4
add	t3,a0,t3	#t3= a0+t3
lw	t4,0(t3)	#loadword
bne	a2,t4,elif	#if a2!=t4 jump to elif
add	a0,x0,t2	# a0=x0+t2
jr	ra	#jump

```

elif:
    bge    a2,t4,else           #if a2>t4 jump to elif
    addi   t1,t2,-1            # t1= t2-1
    j while                     #jump
else:
    addi   t0,t2,1              #t0=t2+1
    j while                     #jump
end:
    addi   a0, x0, -1
    jr ra

```

Section II

Code by Suraj Kumar

Comment by Diya Jain

Ques 1:

```

A.      add t0, zero, zero      #t0 = null
      Do:                      #do statement
          slli t1, t0, 2         #t1= t0 *2
          add t1, a0, t1        #putting address of array
          lw t2, 0(t1)          #loadword in t2 from offset(t1)
          slli t2, t2, 7         #t2= t2*2^7
          sw t2, 0(t1)          #storeword in t2 from offset(t1)
          addi t0, t0, 1         #t0= t0+1

      While: blt t0, a1, Do
      Exit:

```

B.

Ques 2:

```

linear_search:
    add t1, x0, x0              #adding zero to i
For:
    bge t1, a1, end             #if t1 >a1 then jump to end
If:
    slli t0, t1, 2               # shift left t0 by the value of t1 to 4
    add t0, t0, a
    lw t2, 0(t0)                #load word t0 to t2
    bne t2, a2, Ls              #t2 not equal to a2 then jump to Ls

```

	add a0, x0, t1	#add t1 and 0 to a0
	jr ra	#caller call
Ls:		
	addi t1, t1, 1	#add 1 to t1
	j for	#jump to for
End:		
	addi a0, x0, -1	
	jr ra	# return address

Ques 3. TernarySearch:

	addi sp, sp, -48	
	sw s0, 44(sp)	
	addi s0, sp, 48	
	sw a0, -36(s0)	
	sw a1, -40(s0)	
	sw a2, -44(s0)	
	sw zero, -20(s0)	
	lw a5, -40(s0)	
	addi a5, a5, -1	
	sw a5, -24(s0)	#while loop
	j L2	
L8:		#calculating left_mid
	lw a4, -24(s0)	
	lw a5, -20(s0)	
	sub a4, a4, a5	
	li a5, 3	
	div a5, a4, a5	
	lw a4, -20(s0)	
	add a5, a4, a5	
	sw a5, -28(s0)	#calculating right_mid
	lw a4, -24(s0)	
	lw a5, -20(s0)	
	sub a4, a4, a5	
	li a5, -3	
	div a5, a4, a5	
	lw a4, -24(s0)	
	add a5, a4, a5	
	sw a5, -32(s0)	#checking name[left_mid]==target
	lw a5, -28(s0)	
	slli a5, a5, 2	
	lw a4, -36(s0)	

```

    add    a5, a4, a5
    lw     a5, 0(a5)
    lw     a4, -44(s0)
    bne    a4, a5, .L3
    lw     a5, -28(s0)
    j      L4
L3:
                                #if name[right_mid]==target
    lw     a5, -32(s0)
    slli   a5, a5, 2
    lw     a4, -36(s0)
    add    a5, a4, a5
    lw     a5, 0(a5)
    lw     a4, -44(s0)
    bne    a4, a5, .L5
    lw     a5, -32(s0)
    j      L4
L5:
                                #if name[left_mid]>target
    lw     a5, -28(s0)
    slli   a5, a5, 2
    lw     a4, -36(s0)
    add    a5, a4, a5
    lw     a5, 0(a5)
    lw     a4, -44(s0)
    bge    a4, a5, .L6
    lw     a5, -28(s0)
    addi   a5, a5, -1
    sw     a5, -24(s0)
    j      L2
L6:
                                #if name[right_mid]<target
    lw     a5, -32(s0)
    slli   a5, a5, 2
    lw     a4, -36(s0)
    add    a5, a4, a5
    lw     a5, 0(a5)
    lw     a4, -44(s0)
    ble    a4, a5, .L7
    lw     a5, -32(s0)
    addi   a5, a5, 1
    sw     a5, -20(s0)
    j      L2
L7:
                                #else statement
    lw     a5, -28(s0)

```

```
    addi a5,a5,1
    sw   a5,-20(s0)
    lw   a5,-32(s0)
    addi a5,a5,-1
    sw   a5,-24(s0)
```

L2:

```
    lw   a4, -20(s0)
    lw   a5, -24(s0)
    ble  a4,a5,.L8
    li   a5,-1
```

L4:

```
    Add  a0, a5, x0
    lw   s0,44(sp)
    addi sp, sp,48
    jr   ra
```

Judgement: the code is correct.